Chris Thomas Varghese

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Summary

Software Development Engineer with experience designing, building, and operating production-grade backend, distributed, and AI-enabled systems from architecture through deployment. Strong in Python and TypeScript application development, REST and WebSocket APIs, event-driven architecture, asynchronous processing, data pipelines, agent orchestration, cloud services, observability, testing, and CI/CD. Delivered clinical intelligence, real-time analytics, automation, and developer-platform products while translating ambiguous requirements into reliable, maintainable services.

Experience

Stance Health 2025 – 2026

AI Systems Engineer *Bengaluru, India*

- **Clinician Agent:** Built a voice-driven documentation system used across a reported 50+ clinics and 200+ concurrent users, with a reported 40% reduction in form-filling time; verify these metrics before publishing.
- **Customer Agent:** Delivered a conversational intake system with a 6-section clinical form flow for guided PROM collection, adaptive question delivery, audio responses, report uploads, resumable sessions, and final clinical summaries.
- **Clinical Auditor, Summarizer & Phase Intelligence:** Unified 3 concurrent report/VALD data streams with scheduled 30-minute processing and a 15-day staleness fallback for patient timelines, phase mismatches, and evidence-based progression insights.
- **Prognosis Classifier:** Built clinical prognosis, triage, recommendation, and patient-dashboard workflows with 8+ backend analysis routes and automated report generation.
- **Athlete Analytics & Exercise Intelligence:** Developed VALD-powered performance reporting for a reported 100+ users and 50% latency reduction, plus computer-vision form analysis targeting 30+ keypoints, 500+ FPS, sub-50 ms latency, and 35% accuracy improvement; verify reported metrics.
- **TeleCRM Agentic Voice System:** Architected lead tracking and voice-calling workflows that address 2,000+ leads per day through stateful orchestration, CRM updates, and an evaluation harness.
- **Retention & Attendance Platforms:** Built 3 retention reporting workflows and an attendance/occupancy platform with 3 deployment modes for clinic operations, follow-up workflows, and centre-level dashboards.
- **Clinical Intake Infrastructure:** Delivered deterministic interview orchestration and a reusable MCP question-bank service with 2 transport modes for adaptive first-visit and follow-up workflows.

Projects

Codeshipp – Multi-Provider Agentic Coding CLI | *TypeScript, MCP, Gemini, VS Code* April 2026 – Present

- Developed a terminal-first multi-provider coding-agent CLI with streaming responses, tool execution, file operations, agents, tasks, MCP servers, cloud/local model routing, VS Code integration, web retrieval, gRPC, and local search.

Safety-Constrained MongoDB MCP Server | *TypeScript, MongoDB, MCP* Ongoing

- Built a safety-conscious MongoDB MCP server with bounded inspection and CRUD tools, schema inference, operator restrictions, read-only validation, destructive-operation gates, response-size limits, BSON handling, and URI redaction.

Replay – Visual Browser Automation Platform | *Next.js, React Flow, Browserbase, Stagehand* 2026 – Present

- Implemented a visual browser-automation workflow editor with typed trigger/action nodes, registry-driven execution, browser execution, collaboration, persistence, authentication, job triggering, and session playback foundations.

BODS Data Extractor – UK Bus Timetable Intelligence Pipeline | *Python, Pandas, GeoPandas, Plotly*Academic proje

- Built a Python Bus Open Data Service pipeline that downloads ZIP/XML TransXChange datasets, extracts service/route/stop records, and supports expired-service analysis, BODS/OTC comparison, timetable generation, CSV export, geospatial processing, and filtering.

LoRaWAN Smart Streetlight Control Platform | *Arduino, LoRaWAN, Python, Firebase* 2024

- Built an IoT streetlight-management prototype connecting Arduino/LoRaWAN devices to Firebase-backed remote monitoring and control workflows; the reported 20% response-time reduction and hackathon status should be verified before publishing.

Skillsync – Agentic Resume Intelligence Platform | *Python, FastAPI, LangChain, React/Vite* Ongoing

- Developed a resume-optimization platform with FastAPI and LangChain workflows for PDF/job analysis, skill-gap analysis, tailored-resume generation, LaTeX compilation, OCR fallback, authentication/usage limits, and a React/Vite interface.

EEG-Based Brain-Computer Interface Prosthetic Arm | *Python, Scikit-Learn, Raspberry Pi* 2024

- Developed a 3D-printed prosthetic-arm prototype combining EEG signal classification, Raspberry Pi control, and mechanical actuation; won the Best Innovation Award at Intel's Equipathon. The reported 85% action accuracy and patent-pending status should be verified before publishing.

Technical Skills

Programming: Python, TypeScript, JavaScript, Node.js
Backend & Distributed Systems: FastAPI, Express, Fastify, REST APIs, WebSockets, gRPC, asynchronous processing, event-driven architecture, queues, retries, stateful workflows, feature flags, shadow-mode testing
AI/ML & Agent Systems: LangGraph, LangChain, Haystack AI, LlamaIndex, Model Context Protocol (MCP), ReAct agents, Gemini, Google Vertex AI, Google Cloud Speech, Claude, OpenAI-compatible APIs, Scikit-Learn, NumPy, Pandas, OpenCV, DINO, SAM 2, Florence-2
Data, APIs & Integrations: MongoDB, Redis, Amazon S3, Google Sheets API, Google Apps Script, Firebase, VALD, GraphQL, Apollo Client, Zod, Pydantic
Frontend & Visualization: React, React/Vite, Next.js, React Flow, Tailwind CSS, shadcn/ui, Radix UI, React Query, React Hook Form, Plotly, Recharts
Cloud, DevOps & Quality: AWS, Docker, GitHub Actions, CI/CD, build validation, smoke testing, type checking, OpenTelemetry, Phoenix, Langfuse, Browserbase, Stagehand, Trigger.dev, Liveblocks, Clerk, Drizzle ORM, Neon, Firecrawl, Orama
Additional Systems: LoRaWAN, Arduino, HLS playback, OAuth, authentication, observability, privacy verification, OpenAPI contracts

Education

University of Liverpool Liverpool, UK
MSc Advanced Data Science and Artificial Intelligence *2025 – Present*

Karunya Institute of Technology and Sciences Coimbatore, India
Bachelor of Technology in Computer Science and Engineering – First Class *2021 – 2025*

BVB Vivekananda Vidya Mandir Pangappara, TVM, India
Senior School Certificate Examination – Grade: 82.8 *2020 – 2021*